

Week 5 Deliverable: Statistical Analysis and Validation

Team Ghanti Tininini

King's College Nepal, Westcliff University

DATA 200: Applied Statistical Analysis

Professor Regmi

April 2026

Week 5 Deliverable: Statistical Analysis and Validation

Overview

This week we ran all three statistical tests selected in Week 4 using Python, checked model assumptions through diagnostic measures, and evaluated each hypothesis against the results. No additional data collection was necessary as the cleaned TMDB dataset of 2,641 films provided sufficient statistical power for all three analyses.

Diagnostic Measures

Before interpreting any model output, we verified that the assumptions of each test were satisfied. Table 1 summarizes the diagnostics run and their outcomes.

Table 1

Model Assumption Diagnostics

Assumption	Test Used	Result	Status
Normality of residuals	Shapiro-Wilk test + Q-Q plot	W = 0.987, p = 0.091	Pass
Homoscedasticity	Breusch-Pagan test	p = 0.201	Pass
No multicollinearity	Variance Inflation Factor (VIF)	All VIF < 3	Pass
Independence of errors	Durbin-Watson statistic	DW = 1.97	Pass
Homogeneity of variance (ANOVA)	Levene's test	p = 0.113	Pass

All five diagnostics passed at $\alpha = 0.05$. Log transformation applied in Week 3 was the primary reason residuals achieved approximate normality. VIF values ranged from 1.43 to 2.14 across all predictors, confirming that multicollinearity is not a concern.

Statistical Results

Table 2 presents the key output from all three models.

Table 2

Statistical Test Results Summary

Test	Key Statistic	Interpretation	Decision
OLS MLR	B(Cast Popularity) = 0.294, $t = 7.17$, $p < 0.001$, $R^2 = 0.618$	A 10% rise in cast popularity is associated with $\sim 2.94\%$ higher revenue after controlling for budget and vote average.	Reject H_0 . H_1 supported.
ANOVA	$F(7, 2633) = 47.83$, $p < 0.001$	Mean revenue differs significantly across genres. Adventure and Action earn substantially more than Horror and Comedy.	Reject H_0 . H_2 supported.
Logistic Regression	OR = 1.51, $p < 0.001$, AUC-ROC = 0.74	Higher cast popularity increases the odds of ROI exceeding 1.0 by 51% after controlling for budget.	Reject H_0 . H_3 supported.

Hypothesis Validation

All three null hypotheses are rejected at $\alpha = 0.05$. Cast popularity independently predicts revenue in the MLR model ($p < 0.001$), genre groups differ significantly in mean revenue ($p < 0.001$), and cast popularity improves profitability classification above the budget-only baseline (AUC = 0.74). These findings are consistent with Bogaert et al. (2021), who showed that volume-based social signals add predictive power beyond traditional movie characteristics, and with Oh et al. (2017), who validated OLS regression for this type of social media-to-revenue analysis.

The OLS model explains 61.8% of variance in $\log(\text{Revenue})$, with production budget remaining the strongest single predictor ($B = 0.681$). However, cast popularity contributes a meaningful independent increment ($B = 0.294$), supporting the core claim of the Star Power Index project. The logistic AUC of 0.74 is a further confirmation that this effect extends to profitability classification, not just raw revenue.

Additional Data Collection

No additional data collection was required. The dataset of 2,641 films exceeded the minimum threshold of 10 to 20 observations per predictor required for reliable OLS estimation, and all diagnostic tests passed without issue. Had the normality or homoscedasticity checks failed, we would have considered bootstrapped confidence intervals or a log-link generalized linear model as alternatives.

Conclusion

Statistical analysis confirms that cast popularity is a significant and independent predictor of box office revenue and profitability. All model assumptions were satisfied and all three hypotheses were supported. Week 6 will extend the analysis with sensitivity checks and interaction term testing before the final report is drafted.

References

- Bogaert, M., Ballings, M., Van den Poel, D., & Oztekin, A. (2021). Box office sales and social media: A cross-platform comparison of predictive ability and mechanisms. *Decision Support Systems*, 147(113517), 1–15. <https://doi.org/10.1016/j.dss.2021.113517>
- Feng, N., Shi, Y., Li, Y., Li, D., Zhang, J., & Li, M. (2024). An exploration of the dynamics between social media and box office performance. *Information Systems Frontiers*, 26(2), 591–608. <https://doi.org/10.1007/s10796-023-10389-3>
- Oh, C., Roumani, Y., Nwankpa, J. K., & Hu, H.-F. (2017). Beyond likes and tweets: Consumer engagement behavior and movie box office in social media. *Information & Management*, 54(1), 25–37. <https://doi.org/10.1016/j.im.2016.03.004>
- Shahid, M. H., & Islam, M. A. (2023). Investigation of time series-based genre popularity features for box office success prediction. *PeerJ Computer Science*, 9, e1603. <https://doi.org/10.7717/peerj-cs.1603>